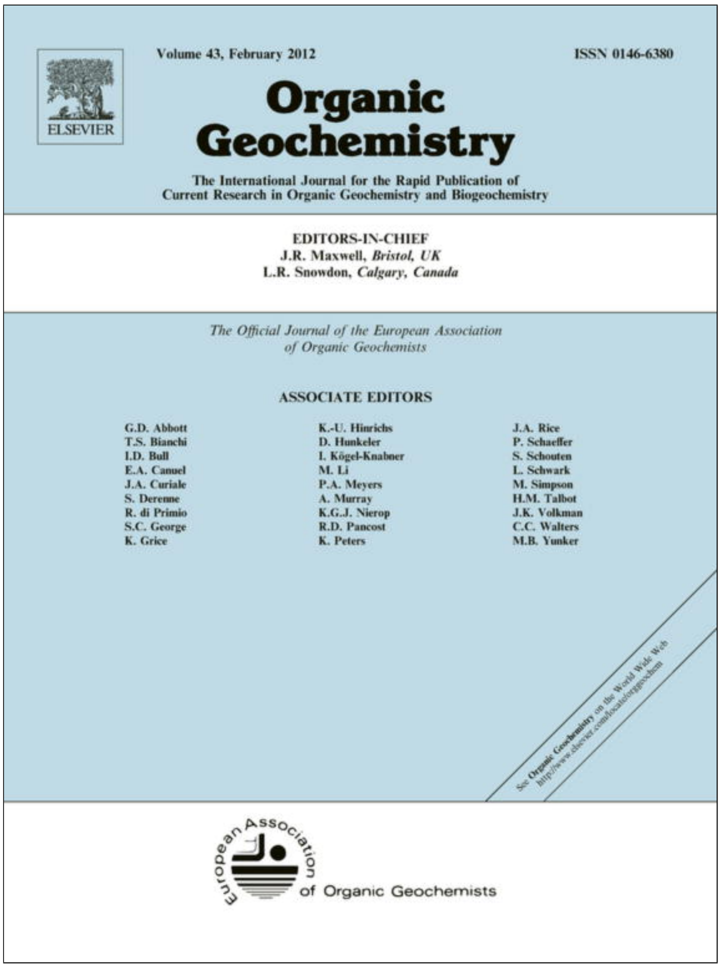
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A practical biodegradation scale for use in reservoir geochemical studies of biodegraded oils

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ABSTRACT

Existing scales widely used to describe the extent of biodegradation of petroleum have insufficient resolution to usefully characterize many heavy oil and bitumen occurrences, including the volumetrically dominant heavily and severely biodegraded oil accumulations in the foreland basins of western Canada and Venezuela. In these and other deposits, existing classifications or descriptions of the biodegradation level may vary only slightly, yet oil may vary in viscosity by orders of magnitude. The “Manco” biodegradation scale proposed here is based on integrating the extent of degradation of various members of compound classes not included in previous biodegradation scales. They include alkyl aromatic and alkyl thiophenic compounds that show variable extent of alteration in samples degraded to uniform levels on standard scales, but which may show variation in local degradation systematics related to biodegradation mechanisms and extent of oil mixing. The Manco scale uses a combination of a consideration of the extent of alteration within a compound class together with a consideration of biodegradation across a range of compound classes. It can be reliably used as a basis for interpreting geochemical changes in heavily biodegraded oil suites and can also be used to differentiate biodegraded oil samples likely to be amenable to cold production from those requiring production strategies such as steam or chemical flooding. As with other biodegradation scales, the scale may also provide evidence for the influx of later, higher quality oil into a reservoir fluid that had been previously biodegraded.

1. Introduction

Heavy oils, super heavy oils and oil sand bitumen are formed by microbial degradation of conventional crude oils over geological timescales. The final distribution of oil API gravity and fluid properties, such as the viscosity of oils found in heavy oil fields, is ultimately controlled by a complex interplay of oil charge rate, pervasive oil charge mixing, microbial diversity and oil chemistry due to compound specific biodegradation rates (controlled by temperature, water chemistry and nutrient supply; Head et al., 2003; Larter et al., 2003, 2006a; Adams et al., 2006). While biodegradation is a primary control on fluid properties in many heavy oil settings such as in western Canada, the nature of the primary oil charge and addition of a secondary fresh oil charge may dominate properties in settings such as in China and western Canada (Koopmans et al., 2002). All these processes complicate existing simple rankings of crude oils in terms of relative level of degradation.

1.1. Previous biodegradation scales

Peters and Moldowan (1993) developed a quasi-systematic classification scheme (PM scale; Fig. 1) to rank the level of biodegradation of an oil on a scale of 1 (least altered) to 10 (most altered). The PM scale has been widely and successfully used, especially for light, conventional oils and condensates. A gas chromatography (GC) trace [or gas chromatography–mass spectrometry (GC–MS) total ion current (TIC) chromatogram] of a non-degraded, conventional crude oil is visually dominated by *n*-alkanes, their presence indicating a pristine or slightly biodegraded oil. Alteration of diasteranes and high molecular weight (MW) aromatic steroid hydrocarbon compounds only becomes apparent when the oil has been very heavily degraded, i.e. at PM 9–10. Another scheme (Wenger et al., 2002) is based on a more comprehensive list of compounds, compound classes and carbon number ranges (Fig. 2), but uses only two categories (“heavy” and “severe” degradation) to describe all stages within the PM 4–10 range. These scales are essentially based on the presence or absence of single key compound classes and a consideration of the extent of alteration within a compound class is accommodated by using dotted or lightly shaded areas in the presence/absence graphs (Fig. 1).

Both these scales present problems in adequately describing the extent of biodegradation in super heavy oils (e.g. western Canada)

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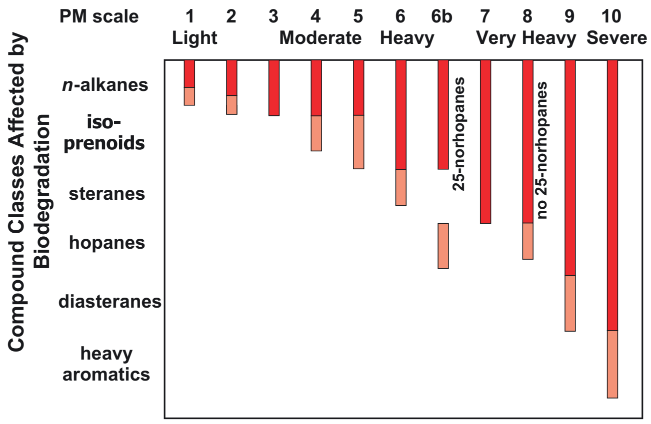

Fig. 1. Biodegradation scale modified after Peters and Moldowan (1993). The lightly shaded areas at the ends of the bars attempt to reflect qualitatively the extent of partial removal of compounds within the class.

because essentially all the heavy oils and Athabasca Tar Sand bitumens fall within PM level 5–8, that is, within the two Wenger categories of "heavy" to "severe" alteration. Peace River oil sand oils are almost all classified at PM level 5, but display orders of magnitude variation in their "dead oil" viscosity values, a property used to evaluate recovery strategies and carry out resource assessments. Knowledge of the extent of degradation is a critical parameter in the western Canada heavy oil belt because the choice of different exploitation strategies that may be applicable depends on the physical properties of the oil (largely viscosity), which correlate strongly with the extent of biodegradation. Applications are not merely practical ones, as biodegradation scales also allow geochemists carrying out more basic study to communicate the relative differences in the extent of the biodegradation process among related samples.

While existing schemes for biodegradation assessment (Volkman et al., 1983; Peters and Moldowan, 1993; Wenger et al., 2002; Peters et al., 2005) have proven useful for conventional light oil, they were generally based on small calibration sets, often from different source facies and basins. Some of the calibration oils had undergone surface evaporation and extensive water washing (Peters and Moldowan, 1993), which preferentially alter an oil in a manner different from biodegradation. They also did not explicitly consider the dominant role of mixing of differently biodegraded oils in the genesis of final fluid composition, nor did they address local variability in biodegradation systematics such as the possibility of aerobic vs. anaerobic degradation (Wardlaw

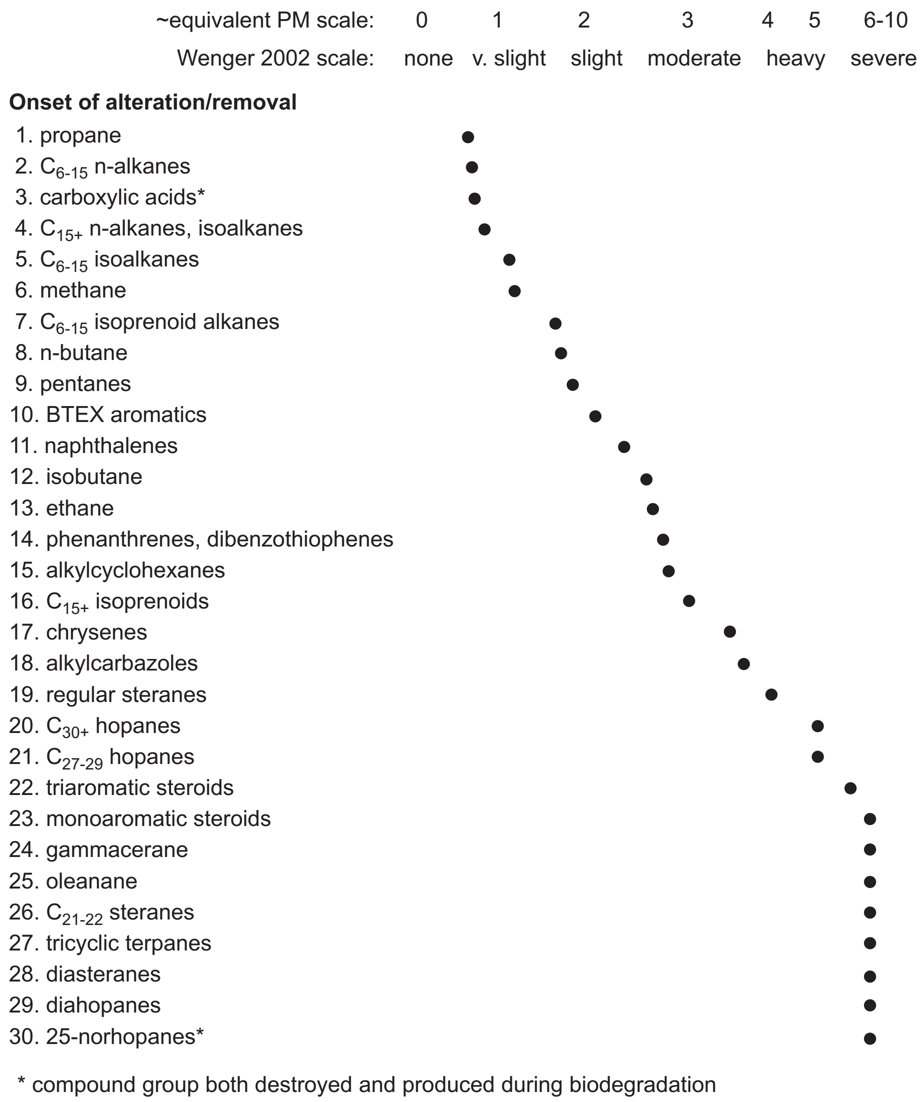

Fig. 2. Removal of selected compound groups at various levels of biodegradation [Head et al., 2003; modified after Wenger et al. (2002)].

et al., 2011) or more likely, different anaerobic degradation pathways (Head et al., 2003). Additionally they did not address the range of variability in oil quality observed at the higher levels of biodegradation common in the volumetrically important oil sands and heavy oil belts of western Canada, Venezuela and elsewhere. In western Canada the oil deposits are commonly biodegraded to a level of PM 5–8 but have highly variable physical properties (e.g. dead oil viscosity varies from thousands to tens of millions centipoises at 20 °C) even though the PM level shows little variation.

1.2. Summary of current views on oil biodegradation

Large scale lateral and vertical variation in biodegraded oil properties within reservoirs are common, due to the interaction of biodegradation and charge mixing (Horstad and Larter, 1997; Larter et al., 2003, 2006a,b, 2008a).

Generation is the rate limiting step in reservoir charging and source rocks typically charge petroleum traps over an interval of the order of a few million to a few tens of millions of years (Pepper, 1991). The notion of a single oil charge to any oilfield is, however, illusory insofar as it could be argued that the oil in any reservoir is an integrated product of various fractions of early, middle and late generation/migration of oil arriving typically over timescales of a million years or longer (Larter et al., 2003). While it is sometimes convenient to consider that the charge arrives episodically, in reality charging is a continuous, if sometimes interrupted, process during source rock expulsion. Thus, by definition, all petroleum accumulations are comprised of mixtures of oils, even before any biodegradation occurs.

Petroleum biodegradation produces compositional variation within oilfields and within individual petroleum columns. Biodegradation proceeds at the oil–water contact (OWC) under anaerobic conditions in any reservoir that has a water leg and has not been heated to >80 °C (Head et al., 2003) and proceeds on a similar timescale to oil charging (Larter et al., 2003). Net degradation of petroleum fractions in reservoirs is primarily controlled by reservoir temperature, the structures of the compounds being degraded and relationship between the OWC surface area and oil volume. Relative volumes of bottom water to oil leg, reservoir water salinity and prior levels of oil biodegradation act as second order controls (Larter et al., 2006b). Typically, degradation flux for fresh petroleum in clastic reservoirs is in the range of 10^{-5}–10^{-3} kg petroleum degraded/m^2 OWC area/yr and rises with decreasing reservoir temperature, from zero near 80 °C, to a maximum flux of <10^{-3} kg petroleum/m^2 OWC/yr at <40 °C (Larter et al., 2003) Nutrient supply from the reservoir aquifer and adjacent shales, mostly buffered by mineral dissolution, probably provides a major control on the range of values of degradation flux.

Biodegradation of all compound classes occurs simultaneously but at different rates (Larter et al., 2006b). That is, the absolute concentration of all of the susceptible compound classes is reduced simultaneously as biodegradation proceeds. However, the *relative* concentration of more resistant compounds and compound classes will increase by virtue of the very rapid loss of mass associated with more labile fractions. The net result of progressive biodegradation *appears* to be a sequential removal of different classes of compounds rather than actual simultaneous removal on different rate trajectories. For light crude oil, the normal removal sequence appears to start with low MW *n*-alkanes followed by higher MW *n*-alkanes, followed in turn by branched alkanes such as the acyclic isoprenoids and then more resistant compounds such as cyclic alkanes. The extent of relative removal generally shows a broad correlation with the physical properties of the oil (e.g. viscosity; Koopmans et al., 2002).

Variation in oil quality controlled dominantly by biodegradation may significantly impact on exploitation strategy. For instance, subsurface biodegradation in a reservoir can produce up to a 50× increase in dead oil viscosity over a distance as short as 40 m within a single reservoir (Larter et al., 2008a), affecting well placement, and higher viscosity oils with dead oil viscosity of many thousands of cP (at 20 °C) may require the use of thermal stimulation using steam or electrical heating to allow production.

Adding fresh oil to an existing biodegraded oil significantly changes key compositional parameters (e.g. relative concentration of biomarkers) but also has a significant impact on the physical properties of the oil mixture. Later oil charges are typically more mature and relatively enriched in gas and light ends relative to the initial charge, so the oil in a subsequent charge effectively acts as a solvent. A relatively small amount of "solvent" (e.g. 5 wt% toluene) added to an oil of 100,000 cP dead oil viscosity will cause a significant reduction in the viscosity of the mixture (typically an order of magnitude) but only a modest increase in API gravity (Larter et al., 2009; Jiang et al., 2010). Changes in the physical properties may be predicted if charge mixing can be quantified accurately or, conversely, the charge history may be determined from a careful consideration of the spatial distribution of the oil properties. The processes and their impact on oil composition and fluid properties have been summarized by Head et al. (2003) and Larter et al. (2006a,b).

The distribution of oil columns subjected to biodegradation is such that basal oil, near the OWC, is usually the most degraded and has the lowest API gravity and the highest polar compound content and viscosity, while the best quality oil occurs near the top of the reservoir (Larter et al., 2008a). Samples from a single oil column in a single well may show vertical variation in apparent biodegradation level that can be dramatic and require careful consideration of the resolution of scale used to describe the extent of biodegradation. Much of the variation in the composition of samples in the middle of an oil column represents vertical mass transport of components downward towards the OWC (Larter et al., 2003), where they are removed through biodegradation. That is, the vertical transport is largely driven by chemical potential (concentration gradients) established by way of the removal of selected compounds through biologically mediated reactions operating at the bottom of the oil leg, near the OWC.

Additionally, or alternatively, compositional gradient may also reflect differential or competing rates of reservoir filling and simultaneous biodegradation during the filling of the reservoir, with commensurate downward or upward migration of the OWC as the reservoir fills, depending on the relative rates of biodegradation and reservoir charging (Larter et al., 2003, 2006b, 2008a,b). The introduction of a second (or subsequent) charge of fresher oil may also be locally constrained (i.e. within only selected compartments in a reservoir) and have an impact on the apparent local extent of biodegradation.

In the Athabasca oil sands, biodegradation has completely removed the *n*-alkanes, branched alkanes and 1–2 ring aromatic hydrocarbons and can affect even the cyclic alkane biomarkers (Bennett et al., 2005). These highly viscous oils are usually rich in asphaltenes and resins (commonly 40 wt% or more). Alkyl aromatic hydrocarbon degradation is one of the best systems for tracking bitumen quality variation in the oil sands. The changes correspond to highly viscous crude oils where dead oil[2] viscosity may exceed 10^6 cP at 20 °C. In the Peace River area, the presence of alkyl naphthalenes and alkyl toluenes indicates less extensive degradation and that the oils may be amenable to cold production with unaltered alkyl naphthalene distributions observed up to viscosity values of ca. 20,000 cP. Extensive alkyl naphthalene alteration and some alkyl

[2] All references to measured viscosity are "dead oil" viscosity at 20 or 22 °C unless otherwise stated. That is, the samples have been degassed naturally during recovery and subsequent exposure to atmospheric pressure and room temperature.

phenanthrene alteration indicate that these oils will most likely need thermal recovery (steam heating) and will not be producible via cold or primary production. Significant alteration of alkyl phenanthrenes and significant increase in the 9-/1-methylphenanthrene abundance ratio suggests that the viscosity of the oil is likely to be well above the cut off for economic cold production in Peace River. In both Athabasca and Peace River oil sand provinces, the PM levels of oils within a single reservoir commonly vary little or not at all, despite the changes in fluid properties.

Many factors, including initial oil composition and oil mixing, may affect the final oil viscosity in a complex manner. Small amounts of light end compounds can disproportionately affect oil viscosity and, in many legacy oil viscosity data sets, loss of light ends from oils during sample storage can easily introduce errors of up to an order of magnitude in viscosity (Adams et al., 2008). It is this sample integrity issue that has prevented effective viscosity-compositional correlation unless specialized oil extraction procedures are applied (Larter et al., 2006b) and sophisticated multivariate methods are utilized for geochemistry-viscosity correlation (Adams et al., 2010). Nevertheless, geochemical indications, even in the absence of calibration oils, can be used to provide general fluid property assessments, so biodegradation level assessment is commonly a routine part of heavy oil reservoir geochemical assessments in applied studies.

1.3. Why do we need a new biodegradation scale?

In general, existing biodegradation scales have insufficient resolution to allow useful characterization of the extent of biodegradation in many heavy oil fields, especially the western Canada heavy oil and bituminous sands belt and the so-called underlying "carbonate triangle" (Creaney et al., 1994 and other papers in the WCSB Atlas). This lack of resolution is limited by the classes of compounds used in the definitions of existing scales. Early scales provided a useful but broad, general level of degradation description across many oil types and had a predominant focus on saturated hydrocarbons, primarily in conventional light oils and condensates. For practical and more fundamental applications, improved resolution was important. For example, in the Peace River oil sands of Alberta, analysis of several hundred samples indicates that, while the biodegradation level ranges from PM 3–6, >90% of the samples would be described by a degradation level of PM 5, even within a single oil column and even though the dead oil viscosity at 20 °C ranges from a few thousand cP to >6 million cP (Adams et al., 2008).

In response to the limitations imposed by existing biodegradation scales, we have developed a higher resolution scale ranging approximately through PM levels 4–8. The "Manco" scale is useful for describing the extent of biodegradation of heavy oil and oil sand bitumen in the Western Canada Sedimentary Basin and unpublished results suggest that it will probably be applicable to other heavy oil and super heavy oil deposits globally. The approach can also be generalized to other oil settings by way of the use of any sequence of progressively less or more biodegradation refractory compound classes to refine the resolution of existing models.

2. Manco scale of biodegradation

In the western Canada oil sands belt with biodegradation level ⩾PM 5, it is typically the concentration and distribution of 1–3 ring aromatic hydrocarbons that most strongly co-vary with the physical properties of the oils. Oil production and upgrading or processing properties are also strongly covariant with these compound classes at intermediate levels of biodegradation, such as at Peace River (Bennett and Larter, 2008). On the other hand, for somewhat shallower reservoirs further to the east at Athabasca, aromatic hydrocarbons with three or more rings and cyclic alkane (naphthenic) biomarkers are more effective for tracking the biodegradation process, oil viscosity and other property variations.

To resolve the limitations of published scales, we have conceived a novel approach using integrated scoring of the biodegradation of several compound classes. The method is different from previous methods in three ways: (i) eight compound classes of differing reactivity, that are appropriate to heavy oil and bitumen, have been identified and used; (ii) the extent of alteration within a compound class has been used in at least a semi-quantitative manner and (iii) the extent of alteration within a compound class and between compound classes has been combined in an algorithm using a linear function within a compound class and a power function between classes. The total hydrocarbon fraction (non-polar fraction) from an oil sample (Bennett and Larter, 2000) was analyzed using selected ion monitoring (SIM) GC–MS for 60 ions at ca. 1.6 Hz with an Agilent 6890N GC instrument coupled to a 5975B mass spectrometry (MS) instrument. ChemStation software was used to control the instrument and to process the data for a range of saturated and aromatic hydrocarbons. Mass chromatograms of various compound classes were inspected and scores ranging from 0 through 4 (*Manco score: Modular Analysis and Numerical Classification of Oils*) were assigned (see below) for each of a suite of eight classes (Table 1). The distributions and scores within the classes change systematically with biodegradation in the range seen for heavy oil and bitumen samples from western Canada (*Manco vector*). A more general classification scheme could use any number of fractions characterized in any manner ordered in terms of generally increasingly refractory behavior. The Manco vector is then algorithmically summed to give a single *Manco number*, which provides an overall estimate of the extent of biodegradation of heavy oil or sand samples. The *Manco scale* is the range of Manco numbers (0–1000) defined for heavy oil and bituminous sand deposits generally relating to PM level 4–8. The maximum value of 1000 is arbitrary but serves to allow resolution of subtly different levels of biodegradation and provide absolute values outside of the range of existing scales, thus avoiding any confusion.

The approach ranks the level of alteration of different compound classes from key mass chromatograms by way of simple visual assessment scored into five levels from 0–4. For the Manco scale, the eight compound classes consist of 1-, 2- and 3-ring aromatic hydrocarbons (i.e. alkyl toluenes, naphthalene + methyl naphthalenes, C_2-, C_3-, C_4-naphthalenes, C_{0-2} phenanthrenes), methyl dibenzothiophenes and steranes. A Manco score of 0–4 may be readily distinguished qualitatively for any compound class from gas chromatograms, as completely non-degraded (Manco score 0) to fully degraded (typically the compounds are absent; Manco score 4), with classes not quite fully degraded (3), or only very slightly degraded (1) and oils mid-way between the extremes (2). This affords five levels of increasing degree of degradation (0–4) for each compound class described as 'pristine', 'light', 'moder-

Table 1
Compounds and classes used as categories reflecting increasing resistance to biodegradation; *m/z* values of the ions used for GC–MS detection are also shown.

Vector element	Compound class	GC–MS *m/z* value
0	Alkyl toluenes	105
1	C_{0-1} naphthalenes (N + MN)	128 + 142
2	C_2 naphthalenes (C2N)	156
3	C_3 naphthalenes (C3N)	170
4	Methyl dibenzothiophenes (MDBT)	198
5	C_4 naphthalenes (C4N)	184
6	C_{0-2} phenanthrenes (C0–2P)	178 + 192 + 206
7	Steranes	217

ate', 'heavy' and 'depleted' in terms of biodegradation for each compound group. Changes in these compound classes have been empirically observed to be applicable to heavy oils, extra heavy oils and bitumen samples collected from the Peace River area and also for much of the Athabasca oil sands as well as samples from other basins. Fig. 3 shows examples of five GC–MS traces for C_{0-2} alkyl phenanthrenes (m/z 178 + 192 + 206 mass chromatograms) with Manco scores from 0–4. Similar descriptions and scores are applied to all eight compound categories in Table 1. The scores are then assembled into an eight element vector (group of numbers) to which is applied an algorithm to calculate a single Manco number used to describe the overall level of biodegradation. A base 5 scale for assessing biodegradation level was chosen by polling several geochemists as to a common, universally agreed generic level of discrimination possible in visual assessment of gas chromatograms of degraded oils. Thus complete removal of a compound class (score 4 – depleted) or no alteration (score 0 – pristine) are easily agreed and it seems mild alteration (score 1 – light) and minor remaining presence of target compounds (score 3 – heavy) are also easily agreed. An intermediate (score 2 – intermediate) represents a mid point in the scale. Attempts at alternate score bases either failed to discriminate the variability seen (bases < 5) or exaggerated visual discrimination possibilities (base > 5). We believe a base 5 system is an optimal and pragmatic scale for visual biodegradation level assessment within a compound class and crucially is equivalent to the n-alkane removal scale used in the PM scale (i.e. 0, intact n-alkanes; 4, n-alkanes gone).

To illustrate the approach, Fig. 4 shows the aromatic hydrocarbon (C_3 alkyl naphthalenes and phenanthrene) concentration in petroleum extracted from reservoir cores, which shows a downwards progressive increase in biodegradation (decreasing compound concentration) to the site of the OWC at ca. 675 m in a well from the Peace River oil sand province (note that the phenanthrene concentration is multiplied by a factor of 10 for visualization purposes). Dead oil viscosity from oils recovered from core samples is also shown for comparison, as are the Manco number 2 (MN2) values (see below for explanation) for the oils and an indication of the PM level which is uniformly 5 throughout the reservoir interval (for visualization purposes this is plotted as PM X 100). There is a general correlation between MN, the hydrocarbon concentration and the oil viscosity. The approach provides added resolution for monitoring the extent of biodegradation, particularly for extensively degraded oils. We note that, although sterane mass chromatograms do not change significantly through the oil column, there is a suggestion that sterane concentration does decrease somewhat at the base of the oil column, indicating fairly homogeneous degradation of steranes. However, the PM scale relates solely to chromatogram-based distribution assessments, so we have maintained the degradation level as PM 5. Below, we discuss the integration of compound concentration and distribution data together in biodegradation assessment and recommend that this is clearly the way forward, but today few labs routinely measure and use compound concentration data. Therefore, we have continued to use visually assessed compound distribution assessments as the primary, rapid biodegradation assessment tool here because that technology is widely available and routinely used.

Table 2 shows the eight categories included in the Manco vector for 71 samples, or plausible theoretical cases, at different levels of biodegradation. The more refractory compound classes are given

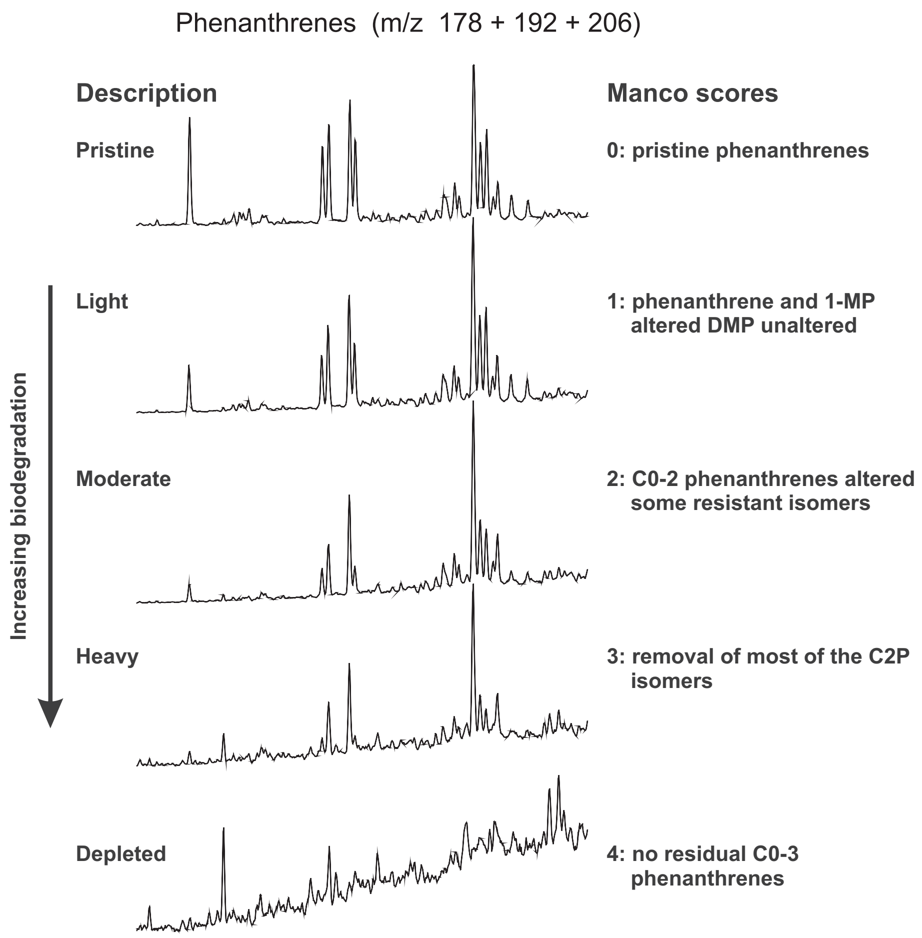

Fig. 3. Examples of GC–MS chromatograms and associated Manco scores for different levels of biodegradation illustrated by the alkyl phenanthrenes.

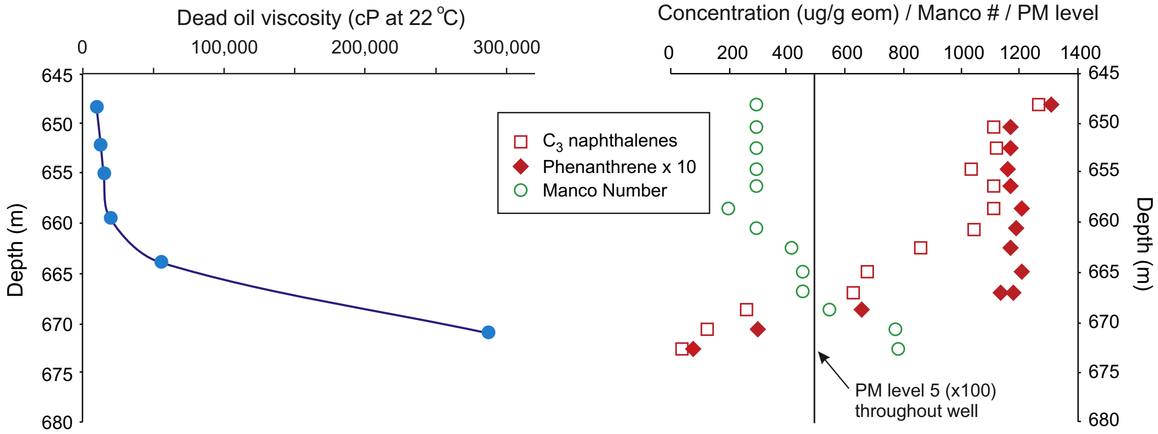

Fig. 4. Aromatic hydrocarbon (C_3 alkyl naphthalenes and phenanthrene) concentration in petroleum extracted from reservoir cores shows a progressive increase in biodegradation extent down to the site of the oil–water contact (OWC) at ca. 675 m in a well from the Peace River oil sand province (note that the phenanthrene concentration is multiplied by a factor of 10 for visualization purposes only). Hydrocarbons diffuse towards the OWC, where they are metabolized by microorganisms using nutrients derived from the water saturated zone below the oil column. Fresh oil is charged to the top of the reservoir, while degradation occurs at the base of the oil column. Compositional gradients reflect this complex charge and degradation scenario, with reactive compounds (e.g. reactive aromatic hydrocarbons decreasing in concentration towards the OWC). Measured viscosity from recovered oil recovered by spinning core samples in a high speed centrifuge is also shown, as are the Manco number 2 (MN2) values and an indication of the PM biodegradation level, which is uniformly 5 (for visualization this is plotted as PM 5 × 100). Note the response of the Manco number and hydrocarbon concentration to biodegradation and the general correlation with oil viscosity.

progressively more weighting in the calculation of the Manco number by observing that the Manco vector (the numerical aggregation of the compound class scores) is calculated as a base 5 number, so a MN value is obtained by multiplying the Manco score (0–4) for each category by 5 raised to the power of the appropriate category number (0–7) for compound classes of increasingly refractory nature. As described above, experiments were carried out with other bases and the apparent optimum balance between the progression of alteration within a compound class (scores) and the progression through the different compound classes was achieved with a base of 5. The use of a simple linear function combining the scores (relative extent of alteration within a compound class) and the observation of any alteration within a compound class (essentially the way the current scales operate) cannot easily accommodate the fact that there is overlap in the alteration of adjacent compound classes. For example, the C_3 naphthalenes are partially altered before the C_2 naphthalenes are completely removed. The sequence of apparent alteration may be somewhat variable because of variable microbial communities, variable oil or water chemistry or a range of other factors including nutrient availability and temperature.

The sum of the compound group scores is termed the MN 1 [Eq. (1)]. The number of compound classes plus log (base 5) of MN1 × the scale maximum minus one (999 in this case), is then divided by the number of compound classes (in this case 8, i.e. 0–7) to yield MN values ranging from 0–1000. This second number (MN2), is the biodegradation level descriptor [Eq. (2)]. Introducing the "number of compound classes" at two points in the calculation returns a value of 1 if MN1 = 1 and makes the approach general such that the numbers of classes of compounds may be varied (see Section 5) to accommodate different circumstances or different compound class combinations.

2.1. Heavy degradation Manco numbers

$$\text{Manco number 1 : MN1} = \Sigma(\text{Class score}_i \times 5^i) \quad \text{for compound classes } i = 0\text{–}7 \qquad (1)$$

The compound classes are listed in Table 1. Scores of the different classes are multiplied by 5^i where i is the class number (0–7). The use of the power function reflects the fact that increasing alteration within a compound class (scores increasing from 0 through 4) is the result of relatively minor differences in extent of degradation, whereas the alteration of an increasingly resistant compound class reflects a very significant and non-linear increment in degradation. Previous biodegradation scales have simply applied a more or less linear category ranking based essentially on the presence or absence of a compound class. The calculation of MN1 for case #2 (base 5 Manco vector = 11000000; written as a base 5 number this would be 00000011) in Table 2 is MN1 = $(1 \times 5^0) + (1 \times 5^1) + (0 \times 5^2) + \cdots = 6$.

$$\begin{aligned}\text{Manco number 2 : MN2} &= [(\text{number of compound classes}) \\ &\quad + (\log_5(\text{MN1}) \times (\text{scale maximum} \\ &\quad - 1)]/(\text{number of compound classes}) \qquad (2)\end{aligned}$$

MN1 = 0 is a special case where MN2 is assigned a value of 0 to accommodate $\log_5$ (0), which is undefined. The number of compound classes is 8 and the arbitrary choice of a maximum MN2 of 1000 was made to avoid confusion between this and existing scales and to allow sufficient resolution at different levels of biodegradation using integer values. The $\log_5$ function is applied to the calculation to return the power function to a more or less linear overall scale while allowing for enhanced weighting of the degradation of more resistant classes of compounds. The spreadsheet formula for evaluating MN2 is thus: MN2 = IF{MN1 > 0 then [8 + (999 ∗ log[MN1,5])] else 0}/8.

The calculation of MN2 for case #2 (MN1 = 6) is thus MN2 = [8 + 999 ∗ $\log_5(6)$]/8 = 140.

Case #26 in Table 2 is an oil with a viscosity of 166,000 cP (20 °C) with a Manco vector of 44310000 (written as a base 5 number this would be 00001344), which indicates that the alkyl toluene and naphthalene + methyl naphthalene classes are completely removed (scores of 4) while the dimethyl naphthalenes are severely altered (score of 3) but are not entirely absent and the trimethyl naphthalenes are only slightly altered (score of 1). The methyl dibenzothiophenes and more resistant compound classes have all been assigned a Manco score of 0, indicating that no alteration is apparent in the chromatograms. Manco number 2 is calculated as 421.

The Manco number is strongly controlled by the extent of alteration of the most resistant compound category that is affected by

Table 2
Manco scores for 71 samples or theoretical cases showing 8 compound categories (i.e. the Manco vector), calculated Manco number and viscosity data (where available). In general, compound classes more susceptible to biodegradation would be expected to be more altered than more resistant classes (i.e. scores decreasing left to right in the vector). Where this is not the case (e.g. cases 8–12, 14, 16, 17, etc.), this may indicate a second charge of fresh oil into a reservoir containing biodegraded oil and/or unusual behavior of the microbiological consortium.

#	Alkyl-tol	N + MN	C2N	C3N	MDBT	C4-N	C0–2 P	Sterane	MN2	Viscosity cP @ 20 °C
1	1	0	0	0	0	0	0	0	1	
2	1	1	0	0	0	0	0	0	140	
3	2	2	0	0	0	0	0	0	194	37600
4	2	2	0	0	0	0	0	0	194	45000
5	2	2	0	0	0	0	0	0	194	65000
6	2	2	0	0	0	0	0	0	194	46500
7	2	2	1	0	0	0	0	0	281	
8	2	3	1	0	0	0	0	0	291	32300
9	2	3	1	0	0	0	0	0	291	33000
10	2	3	1	0	0	0	0	0	291	35000
11	2	3	1	0	0	0	0	0	291	77000
12	2	3	1	0	0	0	0	0	291	
13	3	3	1	0	0	0	0	0	293	184000
14	3	4	2	0	0	0	0	0	334	75000
15	4	4	2	0	0	0	0	0	335	
16	2	3	1	1	0	0	0	0	398	48000
17	2	3	2	1	0	0	0	0	409	43000
18	3	3	2	1	0	0	0	0	409	73000
19	3	3	2	1	0	0	0	0	409	74500
20	3	3	2	1	0	0	0	0	409	90000
21	3	3	2	1	0	0	0	0	409	103000
22	3	3	2	1	0	0	0	0	409	216000
23	3	4	2	1	0	0	0	0	411	
24	4	4	2	1	0	0	0	0	412	
25	3	3	3	1	0	0	0	0	419	112000
26	4	4	3	1	0	0	0	0	421	166000
27	4	4	3	2	0	0	0	0	455	
28	4	4	4	3	0	0	0	0	483	758000
29	3	3	2	1	1	0	0	0	521	
30	3	4	3	1	1	0	0	0	524	93600
31	4	4	3	1	1	0	0	0	524	94000
32	4	4	3	1	1	0	0	0	524	115000
33	4	4	3	1	1	0	0	0	524	109000
34	4	4	3	1	1	0	0	0	524	129000
35	3	3	3	2	1	0	0	0	534	160000
36	4	4	4	2	1	0	0	0	537	
37	4	4	4	2	2	0	0	0	575	
38	4	4	4	2	3	0	0	0	600	
39	4	4	4	3	3	0	0	0	604	99000
40	4	4	4	4	3	0	0	0	608	136000
41	4	4	4	3	1	1	0	0	649	
42	4	4	4	4	1	1	0	0	651	1E + 06
43	4	4	4	4	3	1	0	0	671	
44	4	4	4	2	0	0	1	0	752	291000
45	4	4	4	2	0	0	1	0	752	396000
46	3	4	3	2	1	0	1	0	755	89600
47	3	4	3	2	1	0	1	0	755	84700
48	4	4	3	2	1	0	1	0	755	137000
49	3	4	4	2	1	0	1	0	755	119000
50	4	3	3	2	2	0	1	0	758	
51	4	4	4	2	2	0	1	0	758	
52	4	4	4	3	2	0	1	0	758	201000
53	4	4	4	3	2	0	1	0	758	197000
54	4	4	4	3	2	0	1	0	758	298000
55	4	4	4	3	2	0	1	0	758	315000
56	4	4	4	3	3	0	1	0	761	327000
57	4	4	4	4	4	0	1	0	764	275000
58	4	4	4	3	2	1	1	0	771	1E + 06
59	4	4	4	4	2	1	1	0	772	
60	4	4	4	3	3	1	1	0	774	
61	4	4	4	4	3	1	1	0	774	170000
62	4	4	4	4	3	1	1	0	774	244000
63	4	4	4	4	3	1	1	0	774	527000
64	4	4	4	3	4	2	1	0	786	628000
65	4	4	4	4	3	1	2	0	817	
66	4	4	4	4	4	2	2	0	824	
67	4	4	4	4	4	3	2	0	830	924000
68	4	4	4	4	3	2	1	1	896	2E + 06
69	4	4	4	4	4	2	2	1	908	4E + 06
70	4	4	4	4	4	3	3	2	954	
71	4	4	4	4	4	4	4	3	983	

biodegradation. For example, samples #50 and #51 in Table 2 both have calculated Manco numbers of 758 despite the fact that for sample #50 the N + MN and C_2 naphthalene categories have lower scores (3) than interpreted values (4) for sample #51. Similarly, sample #52 has a Manco number of 758 despite having a higher Manco score for the C_3 naphthalenes than #51. Clearly, the Manco score of '1' for the C0–2 phenanthrenes dominates the calculated Manco number for all three samples. Thus the Manco number provides information on the most biodegraded phase of a mixed sample, as is the case for other biodegradation scales. Mixing or other unusual processes are brought to light by investigating the details of the less resistant compound categories within the Manco vector (Table 2; see below). Combining absolute component concentration data and the Manco numbers permits more detailed investigation of oil mixing phenomena as discussed below. The MN2 number has proved to be helpful in mapping bitumen quality for oil sand recovery in western Canada and for detecting changes in biodegradation style between areas.

The Manco scale effectively provides a means of discussing biodegradation levels within PM 5–8, or even just within the large range in oil composition found at PM 5 where much alkyl aromatic hydrocarbon and alkyl thioaromatic degradation takes place. In addition to the calculated Manco number for a number of examples with different Manco vectors, Table 2 also includes examples of viscosity values measured at 20 °C (extrapolated to 20 °C for samples #64 and #67). Fig. 5 shows a cross plot of $\log_{10}$ (viscosity) vs. MN2. Typically oils with a viscosity <100,000 cP have a Manco number 2 (MN2) of <500. The low viscosity oil samples with a Manco number of 755 (#46 and #47) are not depleted in alkyl toluenes or C_2 naphthalenes but have suffered degradation of the C_{0-2} phenanthrenes, suggesting that these samples may have had later input of petroleum containing alkyl benzenes that effectively reduced the viscosity of an earlier biodegraded and more viscous oil. These samples plot below the general trend between Manco number and viscosity (Fig. 5). Sample #28 has an unusually high viscosity (758,000 cP @ 20 °C) for its Manco number of 483. This character is associated with a very sharp transition between almost complete degradation of the C_3 naphthalenes (Manco score 3) and pristine methyl dibenzothiophenes (Manco score 0). Other samples with anomalously high oil viscosity values also show a sharp transition in the level of degradation with increasing compound category number (Fig. 5). There is no obvious process or genetic cause that is apparent for these observations and this may be an artefact of the process or may indicate a key observation of process behavior.

3. Manco number interpretation

The Manco *number* (MN2) is derived from a Manco *vector* of Manco *scores* for a series of progressively more resistant compounds or compound classes. Irregularities in the Manco vector (apparent out of sequence degradation trends) are sometimes observed whereby a normally more labile compound class (e.g. low MW *n*-alkanes or alkyl toluenes) is present in an oil where more refractory compound classes such as methyl naphthalenes or methyl phenanthrenes have been selectively altered or removed by biodegradation. Two or more charging source rocks or mixing of two oil charge volumes could yield an "out of sequence" Manco vector, but it could also potentially be the result of unusual oil chemistry and/or an unusual biological community responsible for a "non-typical" biodegradation process. Anomalous oil chemistry or biological consortia are expected to be uncommon and Occam's razor and the ubiquitous occurrence of oil mixing suggests that out of sequence distributions would normally be interpreted to indicate that a second influx of non-biodegraded oil had been added to a reservoir containing previously biodegraded oil. Subsequent biodegradation of the mixture may ensue as well. Well constrained Manco vectors from a wide variety of geographic and stratigraphic occurrences of heavy oil and bituminous sands may, however, provide sufficient empirical observations to help isolate heretofore unknown controls on the progression of subsurface biodegradation. Clearly the scatter in the relationship between viscosity and Manco number (Fig. 5) suggests that there is not a simple relationship between these parameters.

Processes other than biodegradation can be important in controlling reservoired oil properties. The initial composition of the oil and any secondary charge, water washing under extreme and unusual conditions, and loss of light ends from a heavy oil could result in a significant increase in viscosity, independent of the level of biodegradation. Thermal cracking of heavy oil or bitumen in a reservoir is not anticipated to be a significant process in biodegraded oil reservoirs (Wilhelms et al., 2001), but any process that results in addition of low MW components that would act as a solvent reduces the observed viscosity. Heavy oil occurrence in reservoirs

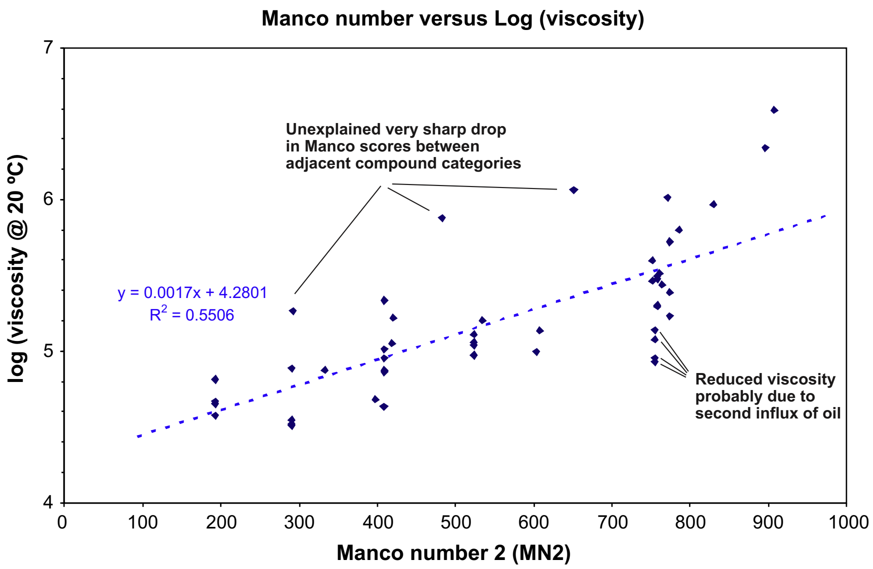

Fig. 5. Relationship between measured viscosity and Manco number 2.

>80 °C is extremely unusual but does occur in complex tectonic settings. The most likely source of light end enriched oils in heavy oil settings is simply later oil charge (Adams et al., 2010). Similarly, natural gas deasphalting caused by the introduction of methane into an undersaturated light oil reservoir could in principle result in a separation of a light, lower viscosity oil from a separated, more viscous, more asphaltene rich oil phase, essentially independent of the extent of biodegradation but, in general, these situations are not commonly observed or expected. Variation in oil charge, including mixing of multiple sourced oils or multiple maturity oil charges, is the principal control on oil viscosity and API gravity for heavy oils, in addition to oil biodegradation.

4. Integration of concentration data and Manco scores to assess processes and oil mixing

Fig. 6 shows the relationship between the absolute concentration (ppm) of alkyl toluenes (AT), C_2 and C_3 alkyl naphthalenes (C2N, C3N), phenanthrene, the 9-methylphenanthrene/1-methyl phenanthrene ratio and the Manco number (MN2) for a large set of contiguous oils. All fractions show a gradual and progressive removal of components (points A, B, C) with degradation (increasing Manco number). Alkyl toluenes are degraded at lower Manco number, but each fraction is degraded continuously over a wide range of Manco numbers, as indicated by decreasing concentration of each fraction. Recent charge is indicated by a high concentration of a component at relatively high Manco number. Methyl phenanthrenes are relatively stable to biodegradation, as seen by the cross plot of 9MP/1MP where alkyl phenanthrene isomer ratios do not change until MN2 reaches ca. 750, but phenanthrene itself is clearly being removed, from Manco number 400, and disappears over a wide range of biodegradation level. As the Manco number is biased towards the presence of the most degraded oil fraction, fresh oil charges are indicated by an anomalously high concentration of components at a given Manco number. The Manco number approach allows visualization of biodegradation trends and assessment of degradation level and is a useful geochemical tool for selection when key isomer ratio changes occur.

5. Classifying oils at PM level 0–4

A similar approach can be used for levels of biodegradation lower than those for heavy oil and oil sand deposits. Here the pertinent classes consist of Class 0 [low MW n-alkanes ($<C_{15}$)], Class 1 [high MW n-alkanes ($>C_{15}$)] and Class 2 (isoprenoid alkanes). Two new Manco numbers are then defined for light and moderate levels of biodegradation as follows.

5.1. Light and moderate biodegradation Manco numbers

Equations for calculating Manco numbers for light to moderately degraded oil are analogous to those shown in Section 2.1 except that only three compound classes are used rather than eight.

$$\text{Manco number 1} : \text{MN1} = \Sigma(\text{Class score}_i \times 5^i) \text{ for classes } i = 0\text{–}2 \tag{3}$$

$$\text{Manco number 2} : \text{MN2} = [(\text{number of compound classes}) + (\log_5(\text{MN1}) \times (\text{scale maximum} - 1)]/(\text{number of compound classes}) \tag{4}$$

If MN1 = 0, then MN2 is assigned a value of 0. The spreadsheet formula for evaluating MN2 is thus: Manco number 2: MN2 = IF{MN1 > 0 then [3 + (999 ∗ log[MN1, 5])] else 0}/3.

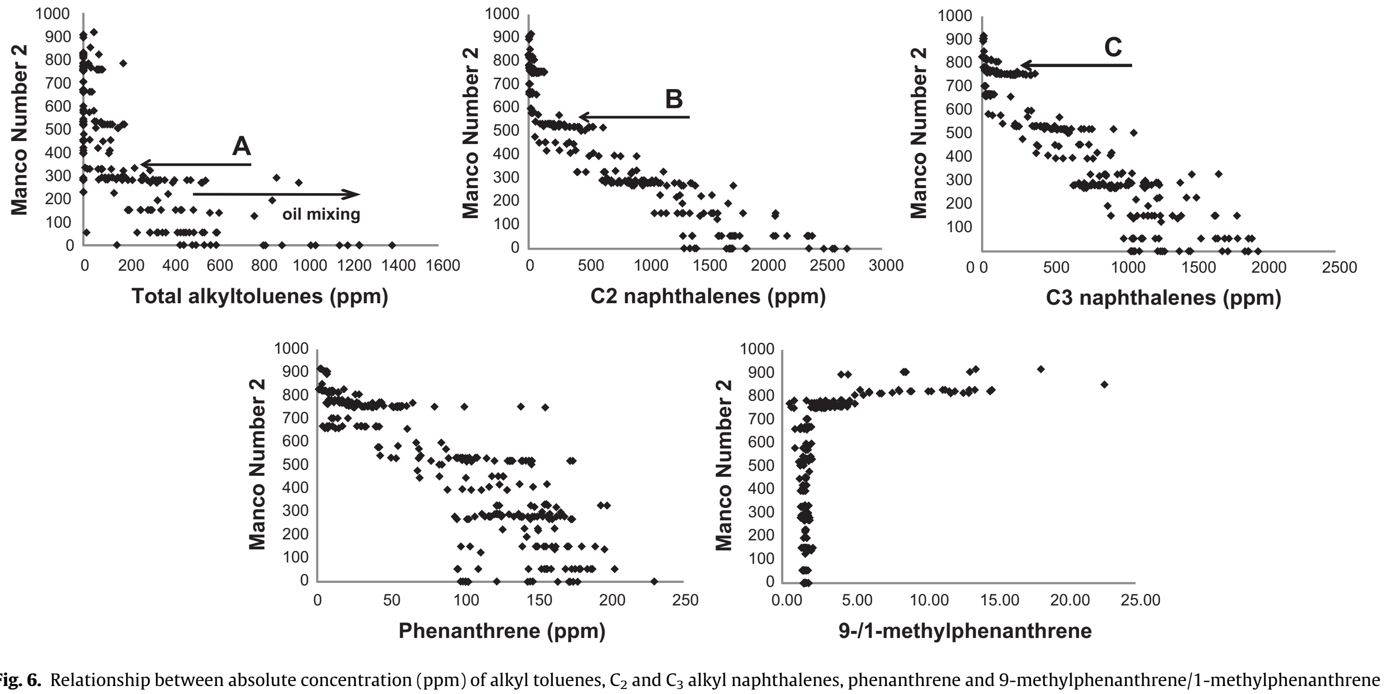

Fig. 6. Relationship between absolute concentration (ppm) of alkyl toluenes, C_2 and C_3 alkyl naphthalenes, phenanthrene and 9-methylphenanthrene/1-methylphenanthrene (9-MP/1-MP) vs. Manco number 2 (MN2). All fractions show a gradual and progressive removal of components (points A, B and C). Alkyl toluenes are degraded at lower MNs, but each fraction is degraded over a wide range of MN, as indicated by the concentration of each fraction. Recent charge is indicated by a high concentration of a component at a relatively high MN. Methyl phenanthrenes are relatively resistant, as seen from the cross plot of 9-MP/1-MP where alkyl phenanthrene isomer ratio values do not change until MN2 reaches ca. 750, but phenanthrene itself is clearly removed by MN 400. As MN is biased to the presence of the most degraded oil fraction, a fresh oil charge is indicated by an anomalously high concentration of components at a given MN [after Adams (2008)]. Note: Data for six related samples from one reservoir compartment are not plotted as they showed anomalous concentrations of *some* aromatic hydrocarbons presumed to be contaminants.

The complete biodegradation level for oils can then be expressed in several ways:

(i) By reporting both light and heavy biodegradation Manco numbers as a couple, e.g. light Manco number, heavy Manco number = 1000, 250 [PM 5] or light Manco number, heavy Manco number = 350, 0 [PM 2]) or light Manco number, heavy Manco number = 500, 450 [PM 5 with light, late charge mixing]; or (ii) by reporting a single combined ultimate Manco number (UMN) by defining a larger biodegradation vector as described below.

This second approach ranks the level of alteration of 11 different compound classes from key mass chromatograms, as before, by simple visual assessment scored into five levels from 0–4. The 11 compound classes consist of class 0 [low MW *n*-alkanes ($<C_{15}$)]; class 1 [high MW *n*-alkanes ($>C_{15}$)]; Class 2 (isoprenoid alkanes); plus the eight classes described above and shown in Table 1, i.e. 1-, 2- and 3-ring aromatic hydrocarbons (alkyl toluenes, alkyl naphthalenes and alkyl phenanthrenes), methyldibenzothiophenes and steranes.

5.2. Ultimate Manco number

Again, the calculations are exactly analogous to those shown above except that the number of compound classes is now 11 rather than 8 or 3.

$$\text{Ultimate Manco number 1} : \text{UMN1} = \Sigma(\text{Class score}_i * 5^i) \quad \text{for classes } i = 0\text{–}10 \tag{5}$$

$$\text{Ultimate Manco number 2} : \text{IF } \{\text{MN1} > 0 \text{ then } [11 + (999 * \log[\text{MN1}, 5])] \text{ else } 0\}/11 \tag{6}$$

The scale can be extended above PM 8 by including hopanes, 25-norhopanes, aromatic steroid hydrocarbons and diamondoid hydrocarbons and so on. While the principle is clear, increasingly complex behavior in that high MW range has been observed and will be reported in a future study. The behavior of non-hydrocarbons is probably more useful in the post PM 8 range.

6. Conclusions

The relative extent of biodegradation of heavy oil and oil sand bitumen may be described in terms of a rigorously defined parameter, the Manco number. The parameter is designed to be useful to detect compositional and thus fluid property differences that are too subtle to be differentiated using earlier biodegradation scales (Peters and Moldowan, 1993; Wenger et al., 2002; Peters et al., 2005). The latter are too insensitive and lack discrimination within the heavy and severe levels of biodegradation commonly seen in heavy oil provinces. The Manco scale is applicable between PM values of 4–8 and can allow recognition of oils containing a mixture of more biodegraded oil and a lighter, typically later oil charge. This new scale adequately differentiates oils near the transition between reservoirs that are cold producible and those for which thermal stimulation is necessary to recover oil. While there is a general correlation between Manco number and viscosity, the relationship is not particularly strong because of factors other than biodegradation that may influence oil viscosity. An expanded ultimate Manco number is also defined which effectively replaces the PM scale from pristine oil (PM 0) up to PM 8.

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